MILITARY SPECIFICATION

COMPRESSOR, POWER DRIVEN, AIR, ELECTRIC MOTOR DRIVEN, 15 CFM, 3,500 PSI, TYPE MB-1

1. SCOPE

1.1 Scope. This specification covers one type of 3,500 psi, 15 cfm, electric motor driven, air or nitrogen compressor, designated type MB-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-V-4719	Valve, Oxygen Line, High Pressure, Aircraft Servicing
MIL-C-83959	Cylinder, Air Purifier
MIL-H-26666	Hose Assembly, Pneumatic, High Pressure
MIL-C-26805	Compressor Units, Air, General Requirements for

DRAWING

Air Force

7347114	Chuck Assembly - High Pressure Air

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

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3.2 General specification. The requirements of MIL-C-26805, as applicable to type III, class 2 compressors, apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Materials.

3.3.1 Valves. With the exception of the servicing valve conforming to MIL-V-4719 (see 3.4.5.2.6 herein), all manually operated valves, i.e., valves used in shutoff, isolation, control service, et cetera, shall be fabricated of corrosion-resistant steel.

3.4 Design and construction.

3.4.1 Compression. Compression from initial to final pressure shall be completed in not less than four separate stages by a reciprocating design.

3.4.2 Unloader. The constant-speed unloader and stop-start control shall be so adjusted that the compressor unit will unload or stop, depending upon which control system is being used, when the pressure reaches 3,500 +175 -0 psi, and will load or start the compressor unit when the pressure drops to 3,200 +150 -0 psi. The upper limit of adjustment of the unloader and stop-start control shall be adjustable over the range of 2,500 to 4,000 psi. The compressor shall automatically unload when stopped so that the compressor unit will always be unloaded upon startup to decrease the load on the drive motor. After the compressor is unloaded, the intercooler and aftercooler separators shall automatically dispose of any condensed moisture, oil, or other foreign matter. No receiver pressure shall be lost during unloading or upon compressor unit shutdown.

3.4.3 Intercooler and aftercooler moisture-and-oil separators. Moisture-and-oil separators shall be provided after each intercooler and the aftercooler to remove all raw moisture and oil. At an ambient temperature of 100° F and a relative humidity of approximately 100 percent, the separators shall be either large enough to permit continuous operation of the compressor unit for a minimum period of 90 minutes without unloading, or automatically capable of draining the separator when required. A separate intercooler and aftercooler moisture-separator valve shall be provided on the primary control panel to permit manual draining of the intercooler and aftercooler moisture separators, the air purifier cylinders conforming to MIL-C-83959, and the filter. An adjustable, spring-loaded-type pressure relief valve, designed and adjusted to operate at a pressure suitable for protection of the compressor, shall be provided on the aftercooler or aftercooler moisture separator.

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3.4.4 Receiver. A receiver designed for an operating pressure of 3,500 psi and having a minimum volume of 500 cubic inches of water shall be provided. The receiver shall be so located that all moisture, oil, or other foreign matter which is separated from the airstream or is drained from the aftercooler will drain toward the receiver drain valve connection. The receiver shall be constructed of ferrous material and shall be easily removable for servicing and testing. It shall be possible to inspect the receiver for fault and corrosion without removal.

3.4.4.1 Receiver drain valve. The receiver drain valve shall be located to insure complete removal of all moisture and oil from the receiver when the valve is opened. The valve shall be operable from the compressor control panel.

3.4.5 Pressure servicing system. The components of the pressure servicing system shall be located as shown on figure 1 and shall consist of a regulator isolation valve, regulator, servicing gage, adjustable relief valve, servicing hose, and servicing valves.

3.4.5.1 Regulator isolation valve. A valve for isolating the regulator from the high-pressure air in the receiver shall be provided.

3.4.5.2 Servicing systems.

3.4.5.2.1 High-low pressure servicing system. The high-low pressure servicing system shall include a servicing hose, regulator, servicing hose pressure gage, regulator isolation valve, servicing valve, and a servicing hose servicing valve.

3.4.5.2.1.1 High-low pressure regulator. A pressure regulator shall be incorporated in the line as shown on figure 1. The pressure regulator shall be suitable for air within the design pressure range and shall be operable without removal of coverings or devices necessary for protection against the environmental or acceleration forces specified herein. The pressure regulator shall perform satisfactorily when tested as specified in 4.2.2.2.5.

3.4.5.2.1.1.1 Working pressure. The pressure regulator shall be designed for a normal working pressure of 3,500 psig and shall meet the flow requirements specified herein.

3.4.5.2.1.1.2 Control knob. The pressure regulator shall be easily adjustable to any outlet pressure setting between 0 and 3,500 psig, inclusive, by means of a single control knob. Not more than 50 lb-in. of torque shall be required to operate the control knob. By counterclockwise rotation of the control knob, the regulator shall bleed the downstream line to any desired lower pressure, including atmospheric, without loss of upstream gas.

3.4.5.2.1.1.3 Overpressure relief device. A differential-type, overpressure relief device shall be provided on the low-pressure side of the regulator. The relief device shall protect the low-pressure side in the event of failure of the regulator. The relief device shall start to vent before the downstream pressure reaches 100 psi above any set dead end pressure up to 3,500 psig, and flow shall reach at least 30 scfm before the downstream pressure reaches 200 psi above the set dead end pressure. The relief device shall reseal after venting when the outlet pressure is reduced to not less than the

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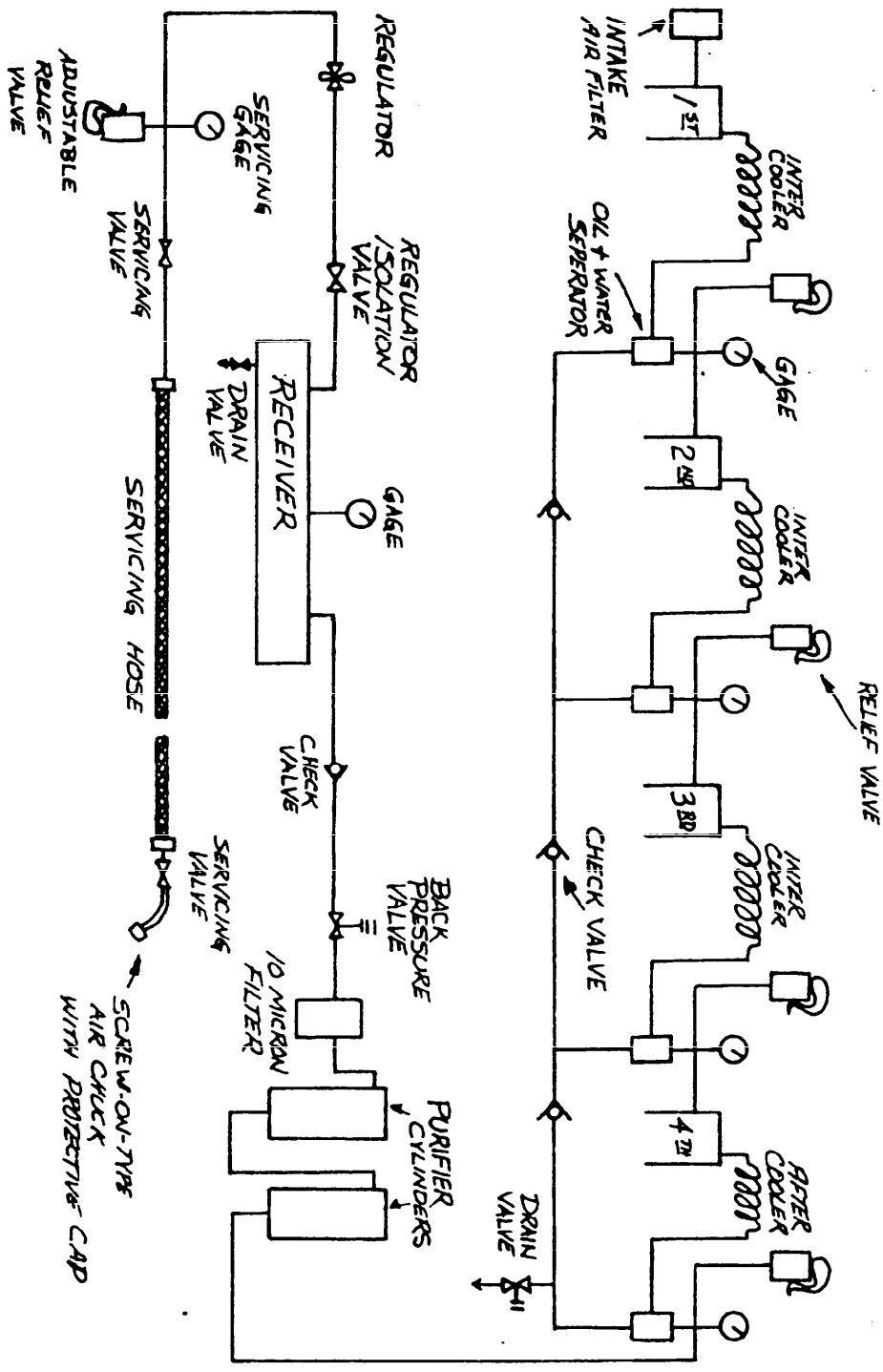


FIG. 1 - COMPONENT LOCATION

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set dead end delivery pressure.

3.4.5.2.2 Screen filter. A screen filter shall be provided at the inlet of the regulator to prevent foreign particles from entering the regulator. The filter openings shall be not larger than those in a 200-mesh screen.

3.4.5.2.3 Servicing hose pressure gage. A servicing hose pressure gage shall be provided. It shall have a minimum face diameter of 4 inches and an accuracy to within 1 percent of full-scale range. The gage shall cover a minimum range of 0 to 4,000 psi with increments of 50 psi. The gage shall give the servicing hose pressure at all times.

3.4.5.2.3.1 Low pressure gage. A second servicing gage shall be provided to indicate the servicing hose pressure for low-pressure pneumatic charging operations. The gage shall cover a range of from 0 to 600 psi, with 5-psi or smaller increments. The gage shall have a dial diameter compatible to the servicing gage, and shall be accurate to within ± 2 percent of full-scale reading. The gage shall be tested at an overpressure of 5,000 psi.

3.4.5.2.4 Air chuck assembly. Unless otherwise specified (see 6.2), a screw-on-type air chuck in accordance with Drawing 7347114-10 shall be provided on the servicing end of the servicing hose.

3.4.5.2.5 Servicing hose. The high-low pressure servicing hose shall be a 30-foot long, size 4 hose conforming to MIL-H-26666.

3.4.5.2.5.1 Hose extension. A 9-inch long, flexible, wirebraided hose extension shall be provided. The hose extension shall withstand a working pressure of 3,500 psi. Adapters shall be used to connect the hose extension to the downstream side of the servicing valve.

3.4.5.2.6 Servicing valves. A valve shall be provided for separating the regulator discharge from the servicing hose to permit setting of the discharge pressure before connecting the servicing hose. A valve conforming to MIL-V-4719, but designed for a working pressure of 3,500 psi and capable of withstanding a hydrostatic pressure of 5,000 psi shall be provided on the servicing hose. The valve shall be capable of withstanding 1,500 opening and closing cycles (see 6.3.1) at 3,500 psi without leakage.

3.4.5.3 Servicing hose rack. A rack or reel shall be provided for storage of the servicing hose. Provisions shall be made for storage of the servicing valve when not in use.

3.4.6 Nitrogen induction. A connection and regulator shall be provided on the compressor for the induction of nitrogen gas. The regulator shall reduce the pressure to some suitable pressure between 40 and 200 psi from storage bottles or other sources having pressures up to 2,215 psi. The required connections, valves, and pressure gages shall be furnished and provisions shall be made to unload any compressor stages not used in the compression of nitrogen. The compressor shall unload when the external nitrogen source pressure drops below 40 psi or any other pressure selected by the manufacturer between 40 and 200 psi. The nitrogen induction design shall be such as to insure that the nitrogen cylinders will be emptied to a remaining pressure of less than 225 psig but not less than 40 psig.

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3.4.7 Mounting. The compressor shall be mounted on a 2-wheel trailer. The two main wheels shall be provided with 6-ply, size 6.00-9 tires and the landing wheel with a 3.50-6 tire.

3.5 Performance.

3.5.1 Rated output. The compressor shall deliver a minimum of 15 scfm of air (760 mm Hg at 70° F) at a minimum pressure of 3,500 psi.

3.5.1.1 Nitrogen gas. The compressor shall deliver a minimum of 15 scfm of nitrogen gas at a minimum pressure of 3,500 psi when supplied with nitrogen through the induction system specified in 3.4.6.

3.5.2 Oil consumption. Oil consumption shall not exceed an average of 1 pint of oil per 25 hours of operation.

3.6 Operational markings.

3.6.1 Warning plate. A warning plate of corrosion-resistant metal and red in color shall be provided in an accessible and visible location near, but not on, the dehydrator assembly. The warning plate shall contain the following instructions:

WARNING: Check purifier cylinders and cartridges for proper operating condition before servicing aircraft.

(1) Change cartridges at intervals indicated on instruction plate.

(2) At time of cartridge change, visually inspect cylinders for presence of oil, water, or other foreign matter. Clean if necessary.

3.6.2 Instruction plate. An instruction plate of corrosion-resistant metal shall be provided and permanently attached to the control panel. The instruction plate shall include the following:

BEFORE SERVICING AIRCRAFT, PRESSURIZE RECEIVER TO AT LEAST 2,500 PSIG AND OPEN RECEIVER DRAIN VALVE. IF MOISTURE IS DISCHARGED, COMPLETELY DRAIN RECEIVER, CLOSE DRAIN VALVE, AND ALLOW SERVICING LINE TO BLEED TO ATMOSPHERE FOR 5 MINUTES BEFORE SERVICING AIRCRAFT. REPEAT 2,500 PSIG PRESSURIZATION CHECK.

3.6.3 Motor input voltage. A corrosion-resistant, reversible, metal plate shall be provided and permanently marked with one input voltage on one side and the alternate voltage on the other side. The plate shall be located under the starter switch and so attached that it may be easily reversed to indicate the nominal voltage.

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3.6.4 Conversion instruction plate. A corrosion-resistant plate marked with complete and readily understandable directions for converting from one voltage to the other shall be permanently mounted on the compressor in an accessible location.

3.7 Dimensions. The overall dimensions of the compressor, including running gear, tow bar, pintle hook, and lunette eye shall not exceed 48 inches in height, 64 inches in width, and 100 inches in length.

3.8 Weight. The dry weight of the compressor shall not exceed 1,800 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The sampling, inspection, and testing of the compressor shall be in accordance with MIL-C-26805 and as specified herein.

4.2 Preproduction testing.

4.2.1 Test samples. Two compressor units shall be subjected to the preproduction tests specified herein and in MIL-C-26805. The preproduction test samples shall be subjected to the following tests as described under test methods of MIL-C-26805. All other tests not listed below may be performed on either unit.

Sample No. 1	Sample No. 2
Examination of product	Examination of product
Operational test	Operational test
Hydrostatic pressure test	Hydrostatic pressure test
Initial teardown inspection	Performance demonstration
Acceleration test	Environmental tests
Performance demonstration	Radio interference suppression test
Endurance test	Mobility tests
Final teardown inspection	Tilted operation
Static pressure	Lifting test

4.2.2 Preproduction tests.

4.2.2.1 Intercooler and aftercooler moisture and oil separator capacity test. The compressor shall be operated for a minimum of 90 minutes at 100° F and a relative humidity of 100 percent without allowing the compressor unit to unload, or shall, where applicable (see 3.4.3), automatically drain the separator when required. Any water carryover from the aftercooler separator into the dehydrator cylinders shall be cause for rejection. If the specified temperature and humidity cannot be obtained, an equivalent test shall be performed by varying the length of the test to obtain an equal amount of raw moisture in the separators. Complete calculations and data shall be included in the preproduction test report giving the amount of moisture entrapped in each separator, size of each separator, and calculated theoretical amount of entrapped water.

4.2.2.2. Pressure servicing system tests.

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4.2.2.2.1 Constant delivery. During the 750-hour portion of the endurance test when the compressor is delivering rated capacity and pressure, the pressure servicing system shall deliver rated capacity at 500 psi discharge with 3,500 psi inlet to the regulator for a minimum of 100 hours.

4.2.2.2.2 Low-temperature test. During start and operation at the 0° F ambient temperature, the pressure-servicing system shall be tested with a 3,500 psi inlet pressure to the regulator. The pressure-servicing system shall be adjusted to deliver rated capacity of the compressor at pressures of 50, 100, 500, 1,000, and 2,000 psi for 10 minutes at each pressure. Capacity measurements shall be taken immediately downstream of the servicing regulator. This test may be performed concurrently with the 2-hour operation of the compressor at rated pressure and capacity.

4.2.2.2.3 High-temperature test. During start and operation at the 125° F ambient temperature, the pressure-servicing system shall deliver rated capacity at pressures of 50, 100, 500, 1,000 and 2,000 psi for 10 minutes at each pressure. The inlet pressure to the regulator shall be maintained at 3,500 psi. Capacity measurements shall be taken immediately downstream of the servicing regulator. This test may be performed during the 4-hour operational test with the compressor delivering rated pressure and capacity.

4.2.2.2.4 Servicing valve. The servicing valve shall be tested in accordance with MIL-V-4719 except that a pressure of 3,500 psi shall be substituted for the 2,200 psi pressure and the valve shall be subjected to 1,500 cycles of endurance testing in lieu of the 500 cycles.

4.2.2.2.5 Pressure regulator. The pressure regulator shall be tested as stated in 4.2.2.2.5.1 through 4.2.2.2.5.5 for compliance with 3.4.5.2.1.1 herein.

4.2.2.2.5.1 Hydrostatic pressure. The high- and low-pressure zones of the regulator, exclusive of the pressure gages and safety relief devices, shall be subjected to hydrostatic pressure of 6,000 psig. After hydrostatic testing, the regulator shall be disassembled and inspected. Any sign of leakage during the test or any indications of permanent deformation to any internal or external component shall be cause for rejection.

4.2.2.2.5.2 Pressure range. With an inlet pressure of 3,600 psig, the regulator shall be set for 0 psig dead-end delivery pressure. With the outlet closed, the outlet pressure shall be increased to 3,500 psig in steps of 25 psi, and then decreased back to 0 psig in steps of 25 psi using the control knob. The torque required on the control knob to change the outlet pressure from 0 to 3,500 to 0 psig shall not exceed 50 lb-in.

4.2.2.2.5.3 Low-flow outlet pressure variation. With an inlet pressure of 3,500 psig to the regulator, the delivery pressure shall be adjusted to 500 psig at a flow rate of 10 scfm. Without further adjustment of the regulator, the inlet pressure shall be slowly reduced to 800 psig. The test shall be repeated with the temperature of the regulator stabilized at +125° F. In no test shall the outlet pressure vary more than ±2 percent throughout the reduction of inlet pressure from 3,500 to 800 psig.

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4.2.2.2.5.4 High-flow outlet pressure variation. The regulator shall be subjected to an inlet pressure of 3,000 psig and the dead-end delivery pressure adjusted to 2,000 psig. With a constant 3,000 psig inlet pressure and no further adjustment of the regulator, a downstream shutoff valve shall be opened until a flow of 75 scfm through the regulator is obtained. The delivery pressure shall not drop to less than 1,900 psig at 75 scfm flow and shall not exceed 2,000 psig when the flow is again reduced to zero. The above test shall be repeated with a constant inlet pressure of 1,500 psig and with an original dead-end delivery pressure of 500 psig. The delivery pressure shall not drop to less than 400 psig at 75 scfm flow, and shall not exceed 500 psig when the flow is again reduced to zero.

4.2.2.2.5.5 Overpressure relief device. With a 3,500 psi inlet pressure, the regulator shall be adjusted for a 500 psig dead-end delivery pressure. Without further adjustment of the regulator, a source of higher pressure gas shall be connected to the regulator outlet and permitted to flow into the low-pressure zone to the regulator. The pressure at which the overpressure relief device starts to vent shall be noted and the flow rate of the effluent gas shall be measured as the outlet pressure increases. The source of higher pressure gas shall be removed from the regulator outlet and the pressure at which the overpressure relief device reseals shall be noted. An initial start-to-vent pressure of more than 600 psig, an effluent gas flow rate of less than 30 scfm at 700 psig, or a resealing pressure of less than 500 psig shall be cause for rejection.

4.2.2.3 Nitrogen regulator and induction test. An external nitrogen source of not less than 2,215 psi shall be connected to the regulator specified in 3.4.6. The compressor shall be operated for 2 hours, delivering a minimum of 15 scfm of free nitrogen at not less than 3,500 psi while the nitrogen supply on the downstream side of the inlet regulator is maintained at the selected pressure (between 40 and 200 psi). Data shall be recorded every 10 minutes in accordance with the performance demonstration test specified in MIL-C-26805 and shall include pressure measurements both upstream and downstream of the inlet regulator. Stopping or unloading of the compressor unit shall be tested by reducing the intake pressure to the compressor below the selected pressure. Failure to discharge nitrogen at the required volume and pressure, or failure to unload when the supply pressure falls below the selected pressure shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The compressor shall be prepared for delivery in accordance with MIL-C-26805. The shipment marking nomenclature shall be as follows:

COMPRESSOR, POWER DRIVEN, AIR, ELECTRIC MOTOR DRIVEN,
15 CFM, 3,500 PSI, TYPE MB-1

6. NOTES

6.1 Intended use. The Type MB-1 compressor is intended for use in charging aircraft pneumatic systems, hydraulic accumulators, oleo struts and tires, or servicing trailers which will in turn service aircraft. The Type MB-1 compressor is suitable for use inside hangars, shops, and other buildings where no winterization provisions are required. Explosion-proof electrical

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components are provided to permit safe use in explosive atmospheres.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a screw-on-type air chuck in accordance with Drawing 7347114 is not to be provided on the servicing end of the servicing hose (see 3.4.5.2.4).

6.3 Definition.

6.3.1 Cycle. A cycle is completed when the valve opens from a closed position to a fully open position, remains open approximately 1 second, and fully closes.

6.4 International interest. Certain provisions of paragraphs 3.4.5.2.5 and 3.5.1 of this specification are subject to international standardization agreement ABC Standard 14/7. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Air Force - 84

Preparing Activity:

Air Force - 84

Project Number:

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